

harmony 2.1.0 on Tahoe-100M: all 95.6 million cells in 4.9 min

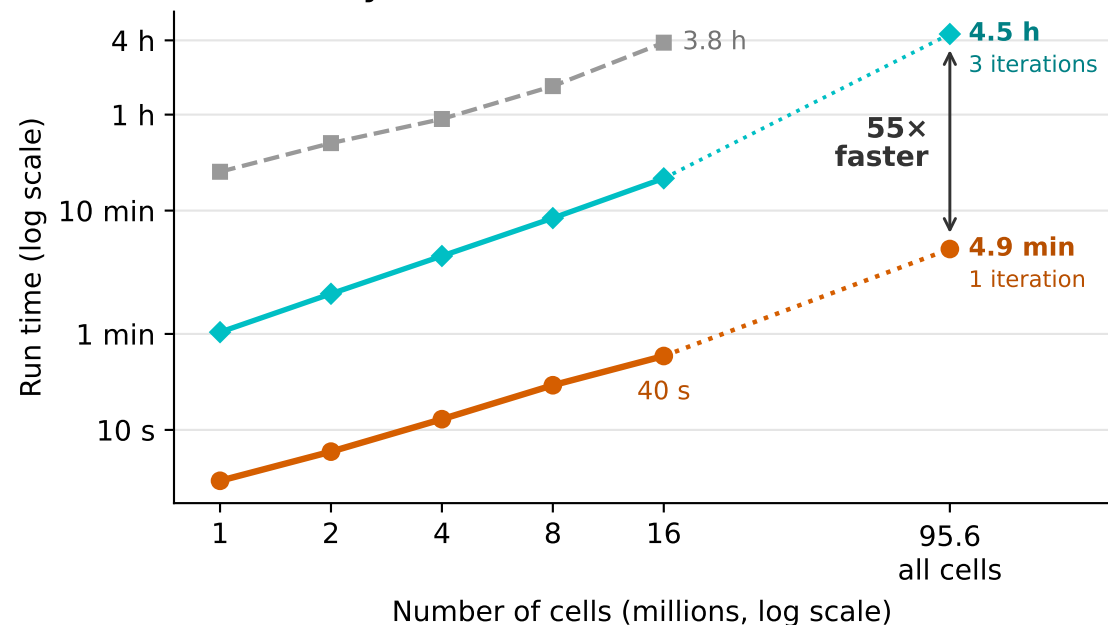
55× faster than 2.0.2 on all cells (4.5 h, where 2.0.2 needed 3 iterations and 2.1.0 1);

28× faster on 16 million cells, with the same number of iterations; 3.3× faster even on one thread (16M cells);

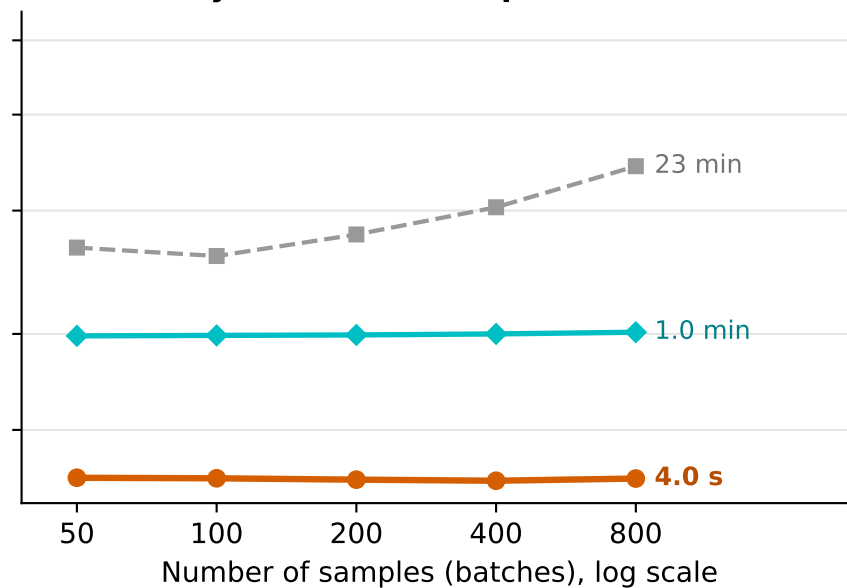
same results as 2.0.2 (per-PC correlation ≥ 0.9999)

—■— harmony 0.2 (PyTorch, April 2026) —◆— harmony 2.0.2 (current release) —●— harmony 2.1.0 (next release)

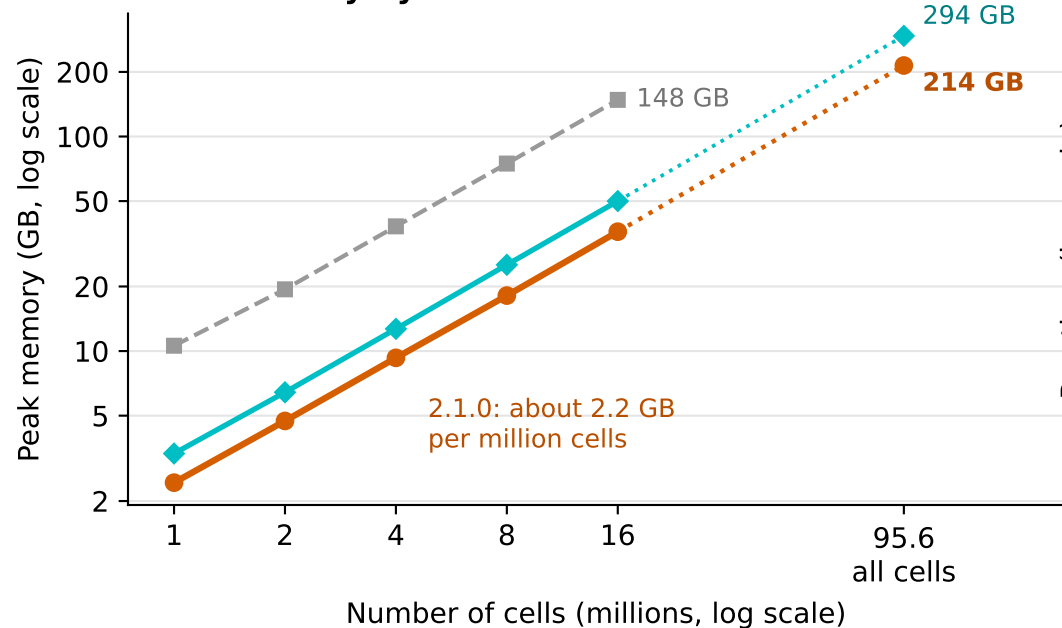
a Run time by number of cells



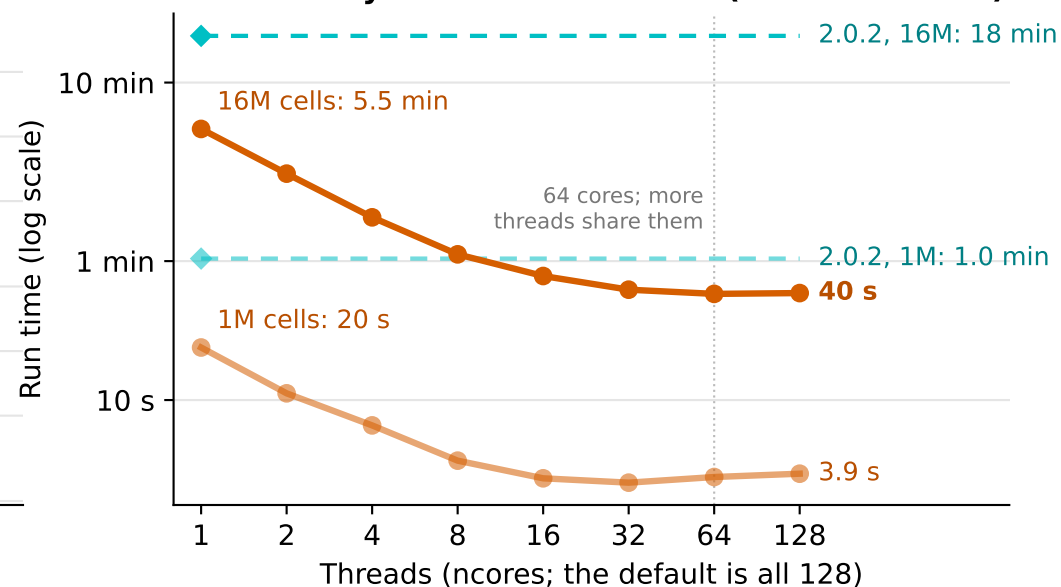
b Run time by number of samples (1M cells)



c Peak memory by number of cells



d Run time by number of threads (2.0.2 uses one)



Data: Tahoe-100M, 50 PCs, corrected for sample; the 1-16 million cell subsamples have 800 samples, all 95.6 million cells have 1,344 (dotted segments). Default settings; for 2.0.2 and 2.1.0, every run converged after 1 Harmony iteration except 2.0.2 on all cells (3). 0.2: April 2026 benchmark on the same machine, not run on all cells; its defaults do more clustering work (epsilon_harmony 1e-4 instead of 1e-2, up to 20 k-means rounds instead of 4), and its memory is what the process held after the run, not its peak. Run time: run_harmony plus reading the corrected coordinates (h.Z_corr), loading excluded (2.1.0: median of 3-5 runs; 2.0.2: one to 3 runs, median). Peak memory: the whole process, input included. Machine: a shared server, 2× AMD EPYC 7543 (64 cores, 128 threads), 3 TB RAM. 2.1.0 (commit 623ff51) used all 128 threads, its default, except in d.